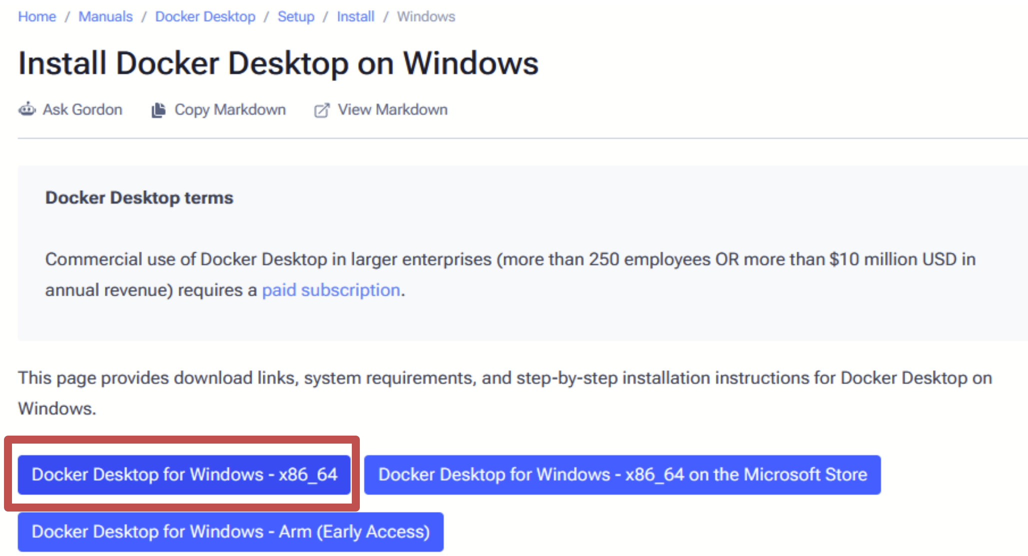
Installation

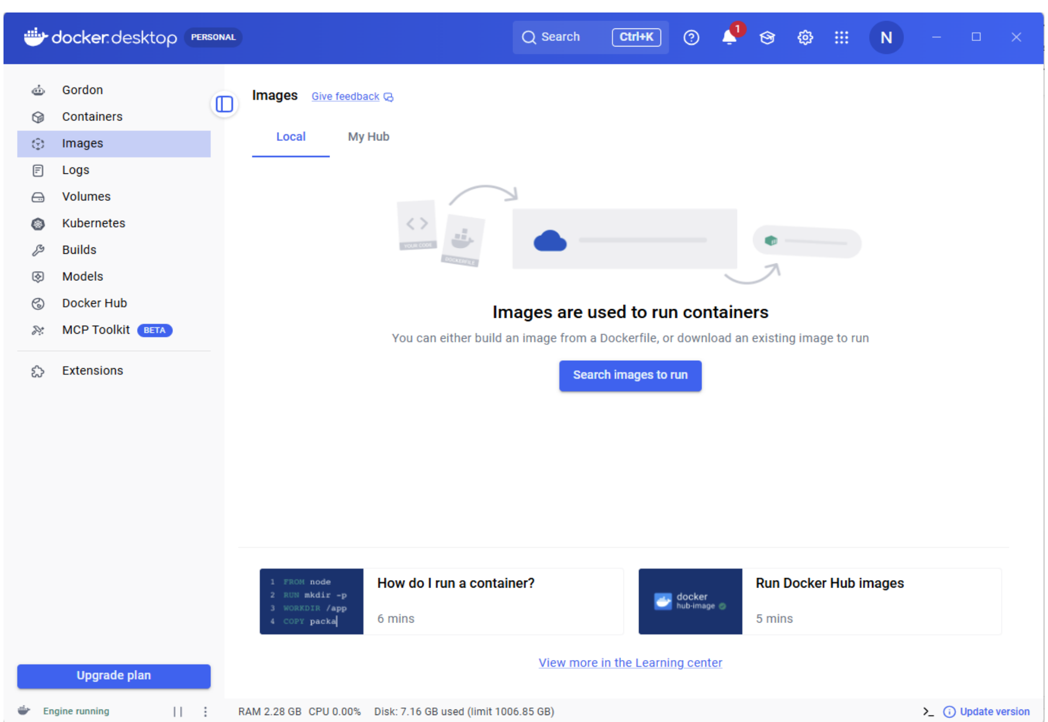
MicroICS is a tool to analyse 3D biofilm images. It runs as a self-contained Docker image (https://hub.docker.com/r/skblaz/microics), accessed through a browser-based interface, and the source code is available on GitHub (https://github.com/SkBlaz/biofilm-classification).

Installation on Windows

- ### Install Docker Desktop
    - On the Docker download page (https://docs.docker.com/desktop/setup/install/windows-install/), click the button for your version of Windows (amd64 or arm64). The download will start.



    - When it finishes, double-click the downloaded file to begin installation.
    - Follow the installer, using the default (recommended) options.
    - Once installed, start Docker Desktop and wait until the whale icon indicates that it is running. Active Docker Desktop looks like this:



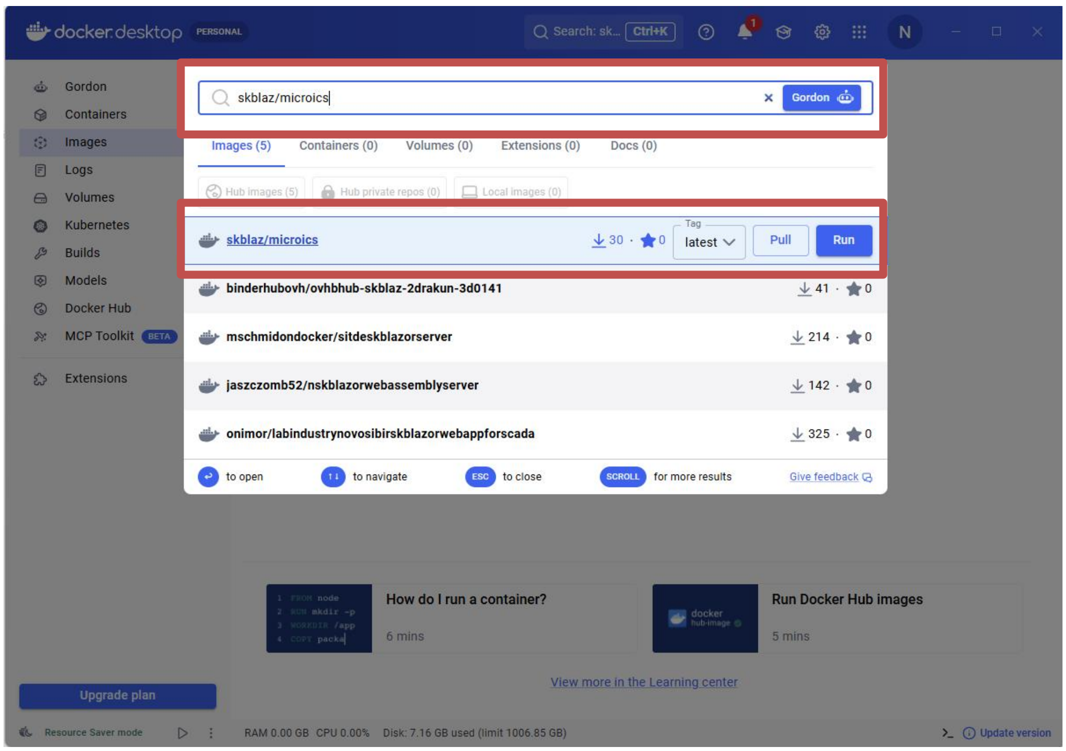
● Setting up MicroICS (via Docker)

You can do this in two ways: using Docker Desktop (the graphical interface – make sure it is running and start it if not), or using the command line (PowerShell on Windows, Terminal on macOS/Linux).

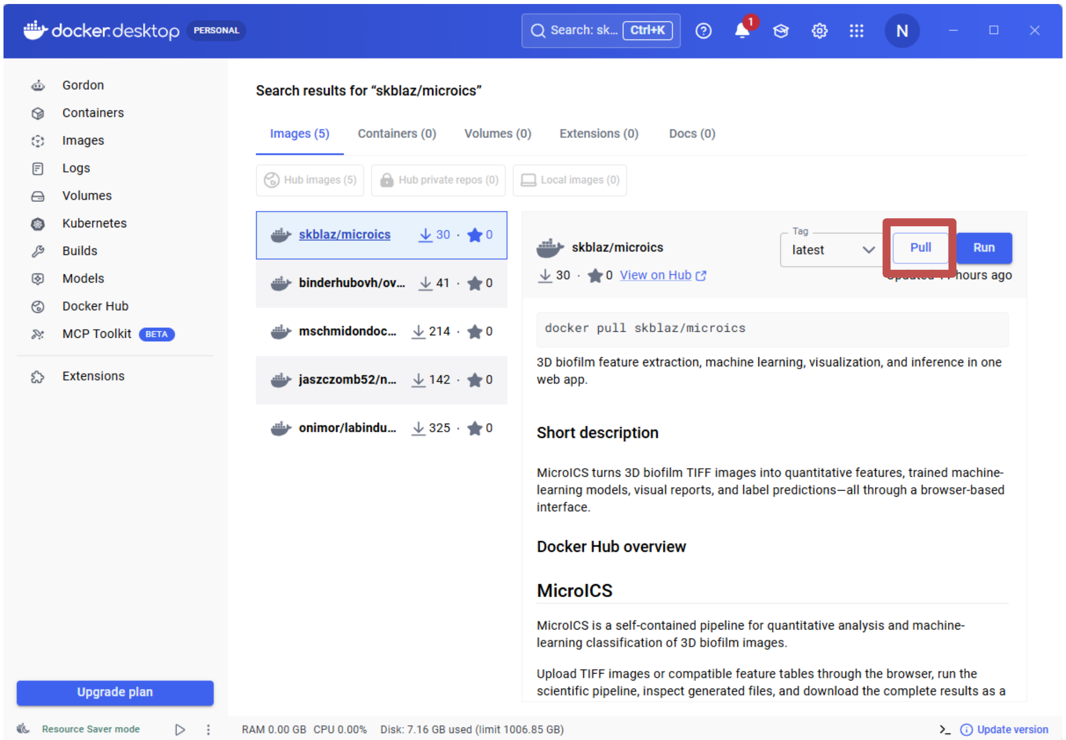
Before Docker is installed and run, WSL, the virtual machine, and Hyper-V may need to be activated on Windows. You can do this by searching for “Windows Features” and selecting “Turn Windows features on or off”. Tick Hyper-V, Virtual Machine Platform, and “Windows Subsystem for Linux”. Click OK and restart the computer, then continue installing Docker. If other issues arise, please resolve them with your IT department.

Option 1 — using Docker Desktop (graphical)

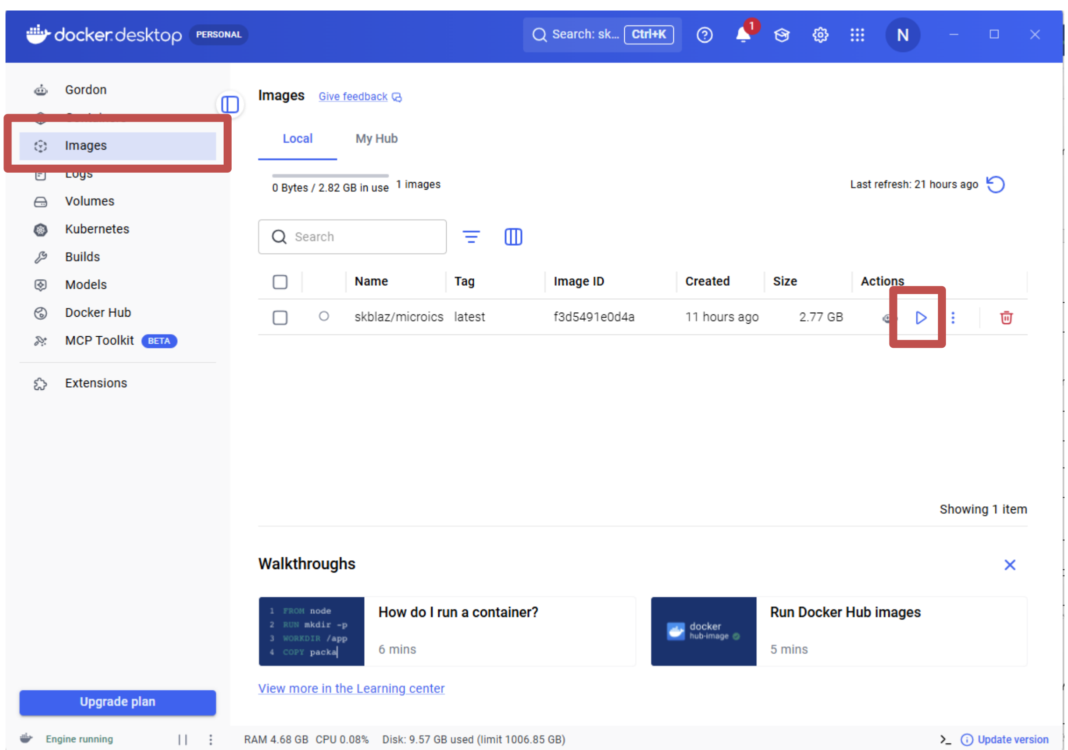
- In the search bar at the top of Docker Desktop, type **skblaz/microics**.



- Select it by clicking, then click **Pull** to download it. This may take a few minutes.

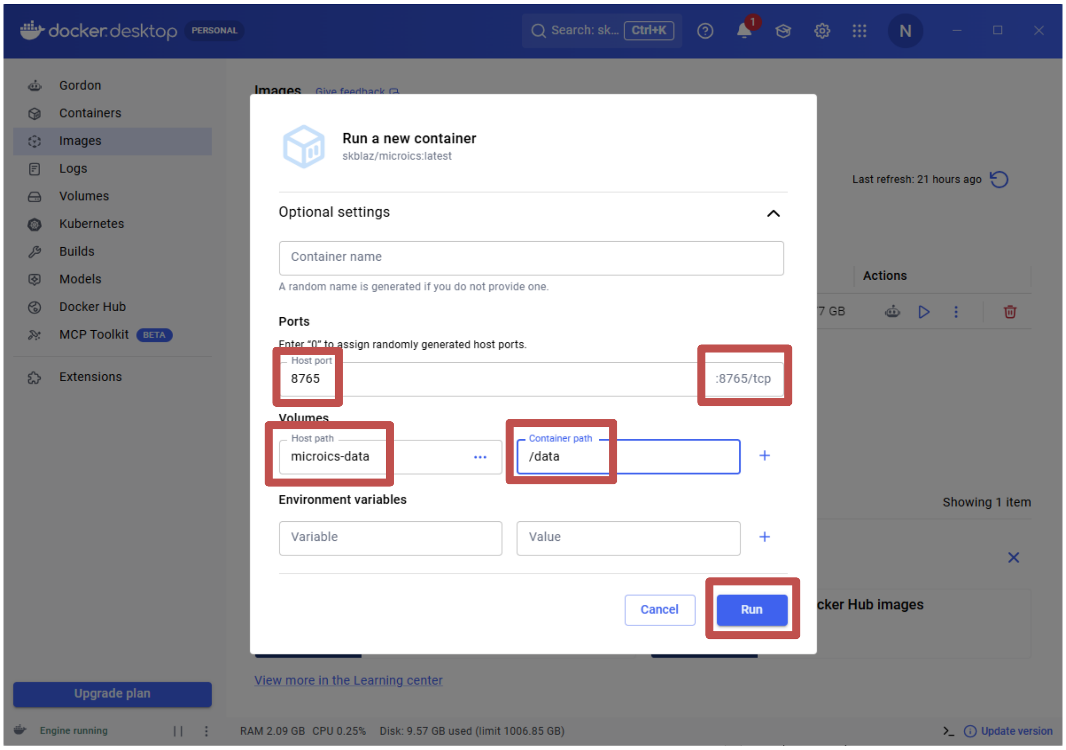


- When it has downloaded, open the **Images** tab on the left. Find skblaz/microics and click **Run** (the ▶ arrow on the right).

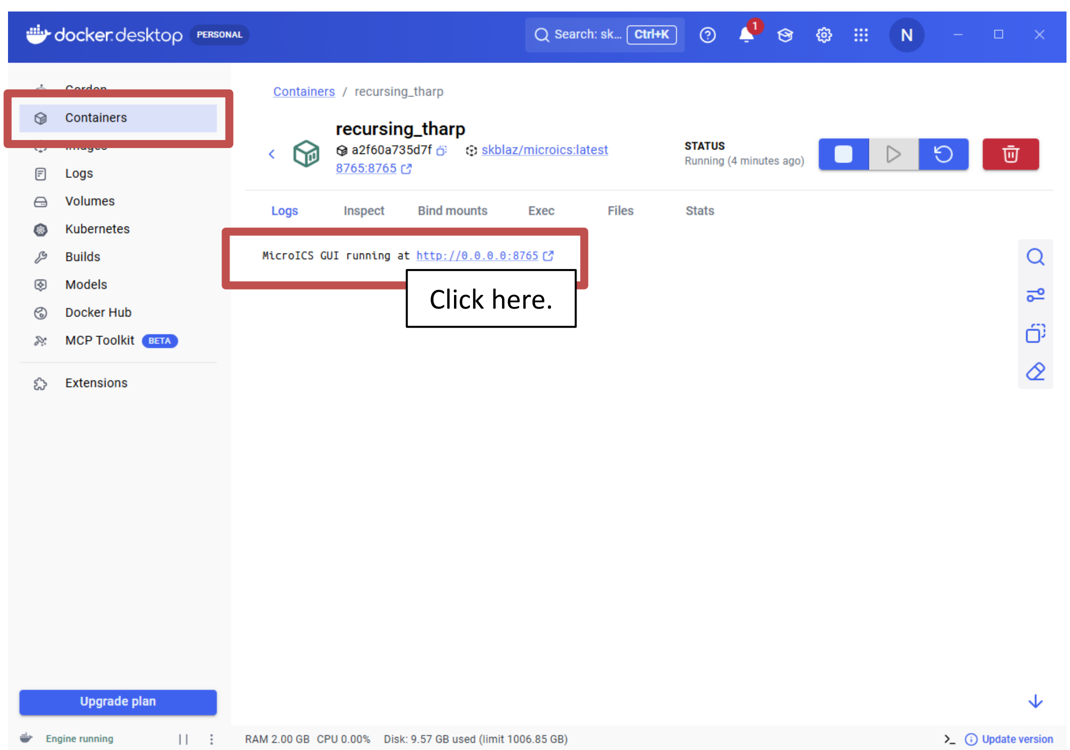


- In the dialogue that appears, click **Optional settings** to expand it, and enter:
  - **Ports** — Host port: 8765 (Container port: 8765, in grey, usually automatically filled)
  - **Host path**: microics-data, **Container path**: /data

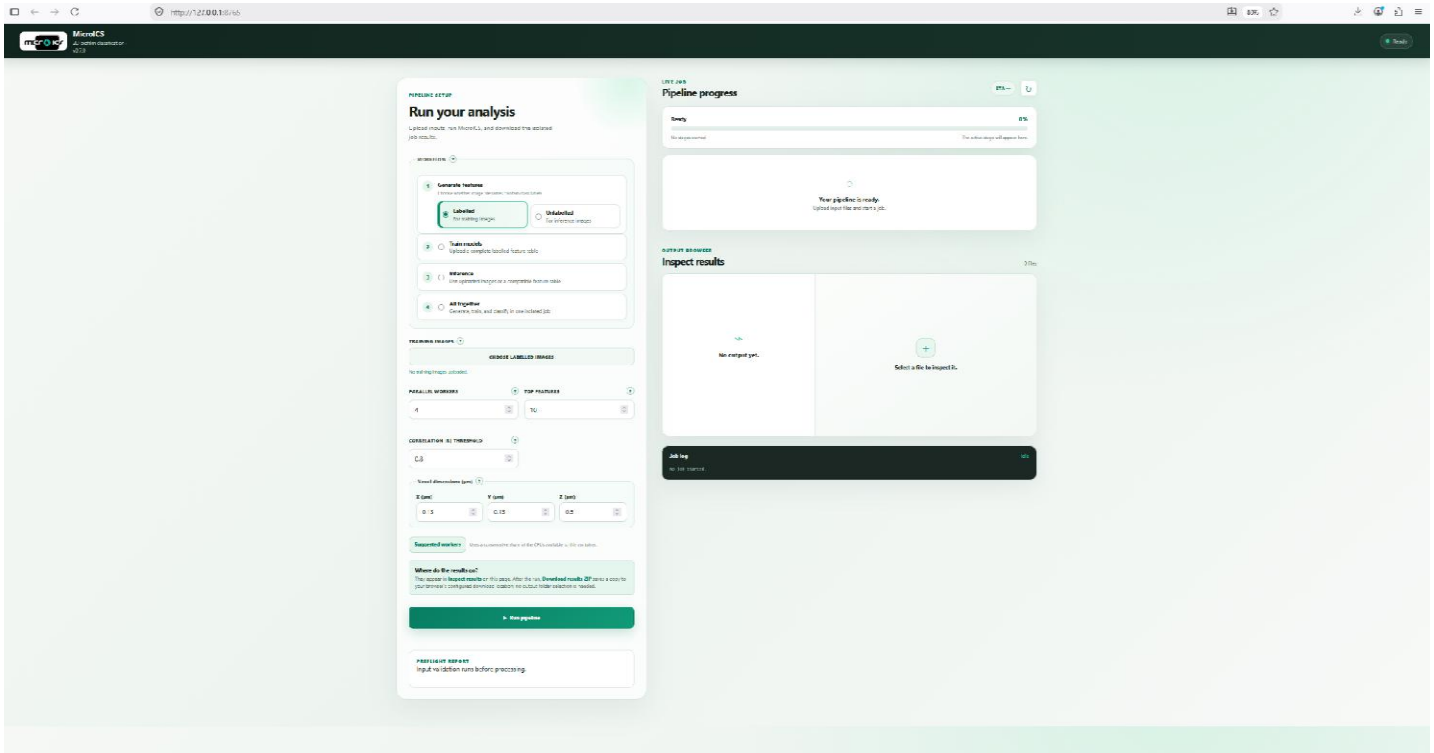
After you enter these, click **Run**.



- The container now appears under the **Containers** tab. Docker Desktop shows a clickable **8765:8765** link next to the running container.



- The MicroICS interface should look like this:



Option 2 — using the command line

- Make sure Docker Desktop is running, then open PowerShell (Windows) or Terminal (Mac/Linux).
- Download the image:

```
docker pull skblaz/microics:latest
```

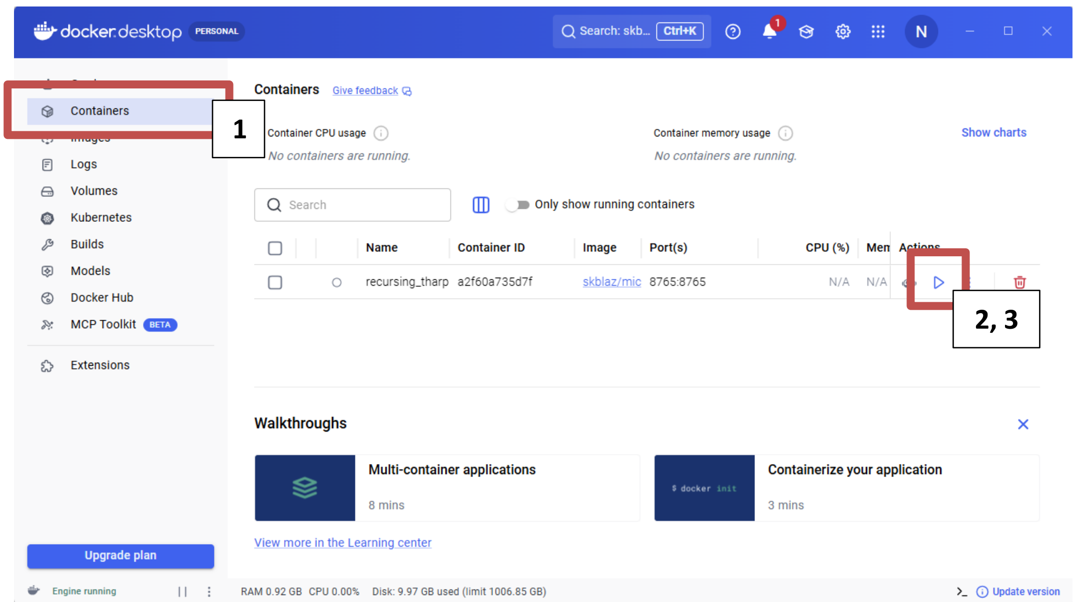
- Start the tool:

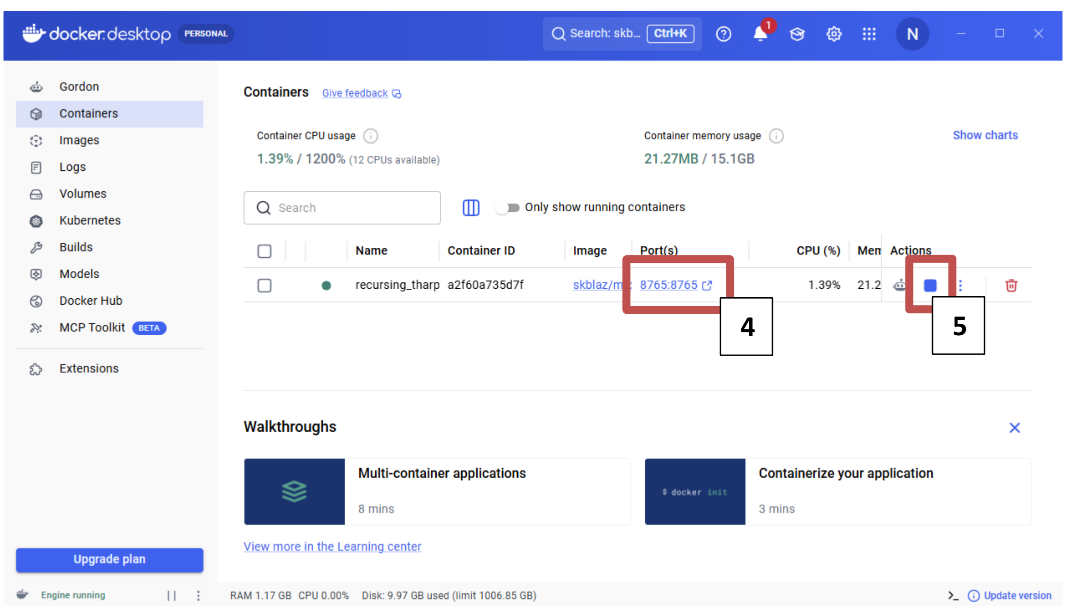
```
docker run -p 8765:8765 -v microics-data:/data skblaz/microics:latest
```

- Open http://localhost:8765 in your browser.
- To stop the container press on (■) below the Actions.

- **Restarting the existing container of MicroICS**
    1. Start Docker Desktop and go to Containers.
    2. Find your MicroICS container (it'll be stopped).
    3. Click the Start button (▶) next to it.
    4. Once it's running, click the 8765:8765 link, or open http://localhost:8765 in your browser.
    5. When you have finished the analyses, stop the container by pressing (■) below Actions.

To get the latest version of the Docker image of MicroICS, you need to repeat the steps for setting up MicroICS (via Docker).





Installation on Linux

- **Install Docker**
    - Download and run Docker's official installation script:

```
curl -fsSL https://get.docker.com -o get-docker.sh
```

```
sudo sh get-docker.sh
```

- Allow Docker to run without sudo:

By default, Docker requires sudo for every command, which causes permission problems with output files. Add your user to the Docker group so Docker can be run directly:

```
sudo usermod -aG docker $USER
```

- Then reboot for this change to take effect:

```
sudo reboot
```

- Check that Docker installed correctly (recommended)

```
docker --version
```

It should print a version number.

Then continue to the "Installing MicroICS with Docker" steps — from that point on, Linux is identical to the Windows (pull the image, run it, open the browser).

Installation on macOS

- **Install Docker**
    - Download Docker Desktop from https://docs.docker.com/desktop/setup/install/mac-install/ and install it. On Mac you may be offered an Apple chip version (M1–M4) or an Intel chip version – check which you have via → About This Mac → look at “Chip” or “Processor”. Install using the default options.
    - Start Docker Desktop: Open Docker Desktop from Applications and wait until the whale icon in the menu bar shows it is running. Docker must be running whenever you use MicroICS.
    - Check Docker is working (optional): Open Terminal and run:

```
docker --version
```

It should print a version number.

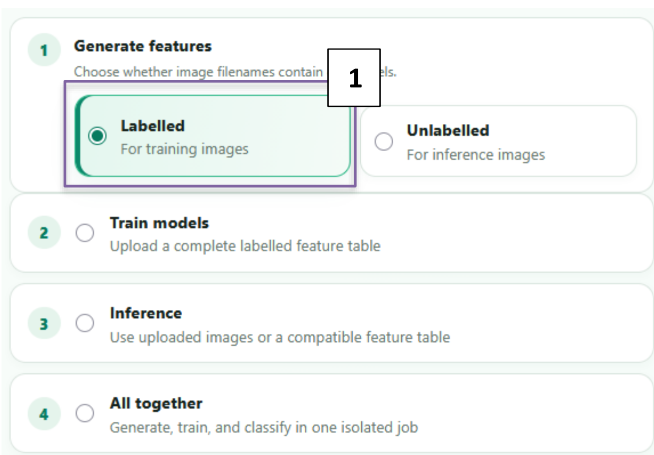
Then continue to the "Installing MicroICS with Docker" steps — from that point on, Mac is identical to the Windows or Linux (pull the image, run it, open the browser).

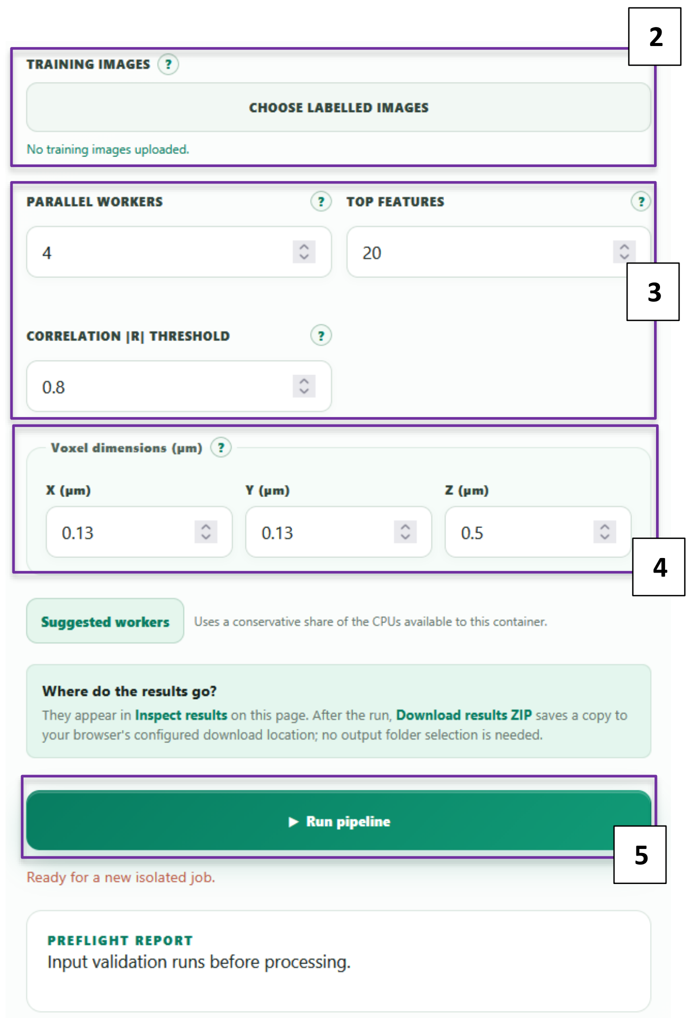
Running the MicroICS

Generate features

1. Choose “Generate features” in mode “Labelled”. Labelled mode extracts the label from each filename, so that the model can link the features to the label during training.
2. Point the software to the folder containing your images in TIFF format, using “Choose labelled images”.
1. Set the running options. “Parallel workers” – how much of your CPU to dedicate to the analysis: use full capacity when you are not otherwise using the computer, and less if you are working on it at the same time. Suggested workers will provide optimal settings for running.
   At this point, “Top features” and “Correlation |R| threshold” do not matter.
3. Adjust the voxel dimensions (the pixel size in micrometres) to match your images.
4. Click “Run pipeline”.

When the run starts, the software also checks the image naming and reports the number of classes it found.

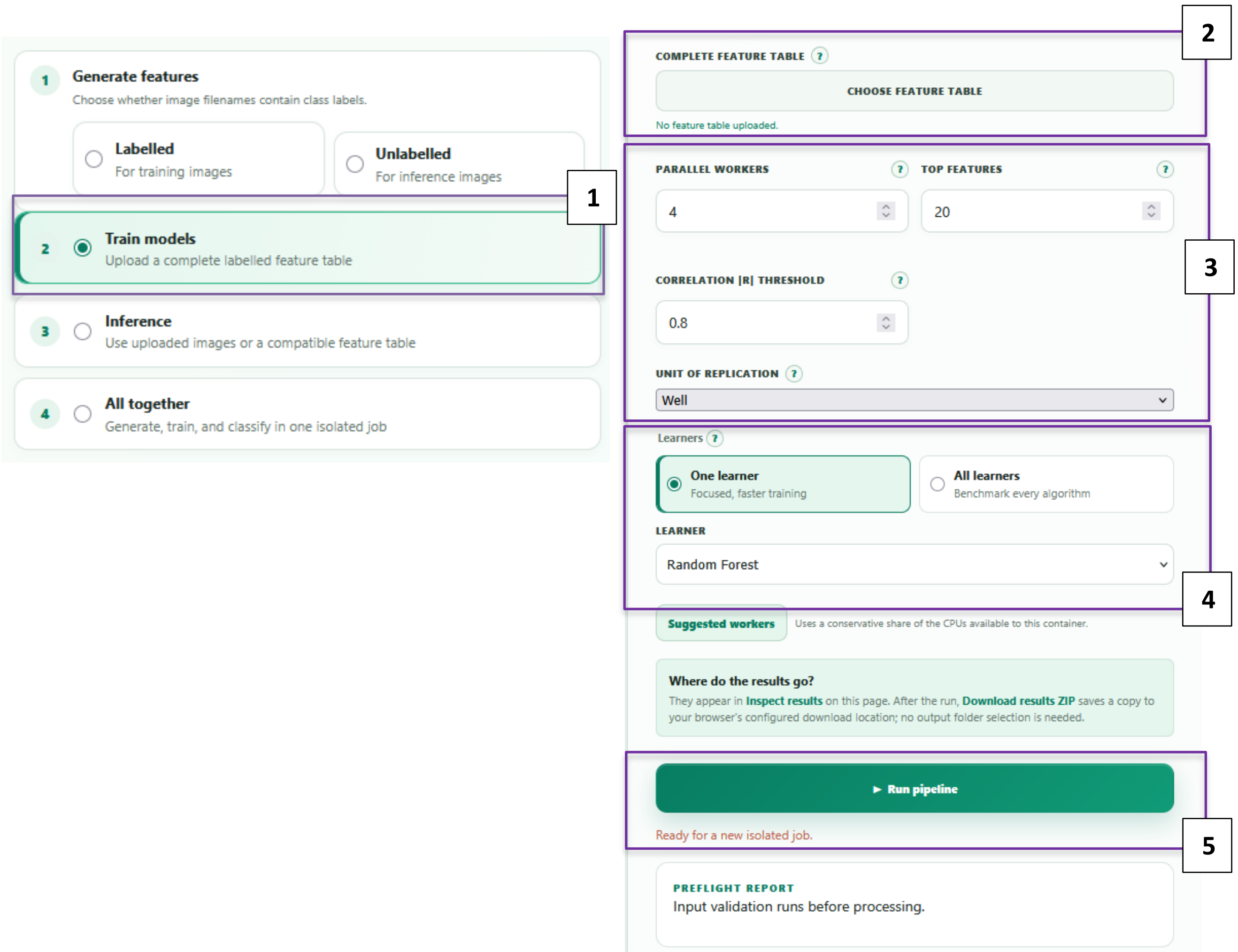




Train a model

1. Choose “Train models.”
2. Navigate to the folder containing calculated features (datafile.tsv) using “Choose feature table”.
3. Set the running options: “Parallel workers” – how much of your CPU to dedicate to the analysis: full capacity when you are not otherwise using the computer, and less if you are working on it at the same time. Suggested workers will provide optimal settings for running.
   “Correlation threshold” – choose how many of the most important features to analyse, and the threshold for the analysis.
   “Unit of replication” – specify what counts as an independent experiment (day, plate, well, or position within the well). This is important for testing the trained model correctly.
4. Choose which learner to train on your data: a single learner and choose learner (model) from dropdown menu, or all those currently available.
5. Click “Run pipeline.”

When the run starts, the software also checks the input datafile for missing values (NaN, empty cells, etc.).



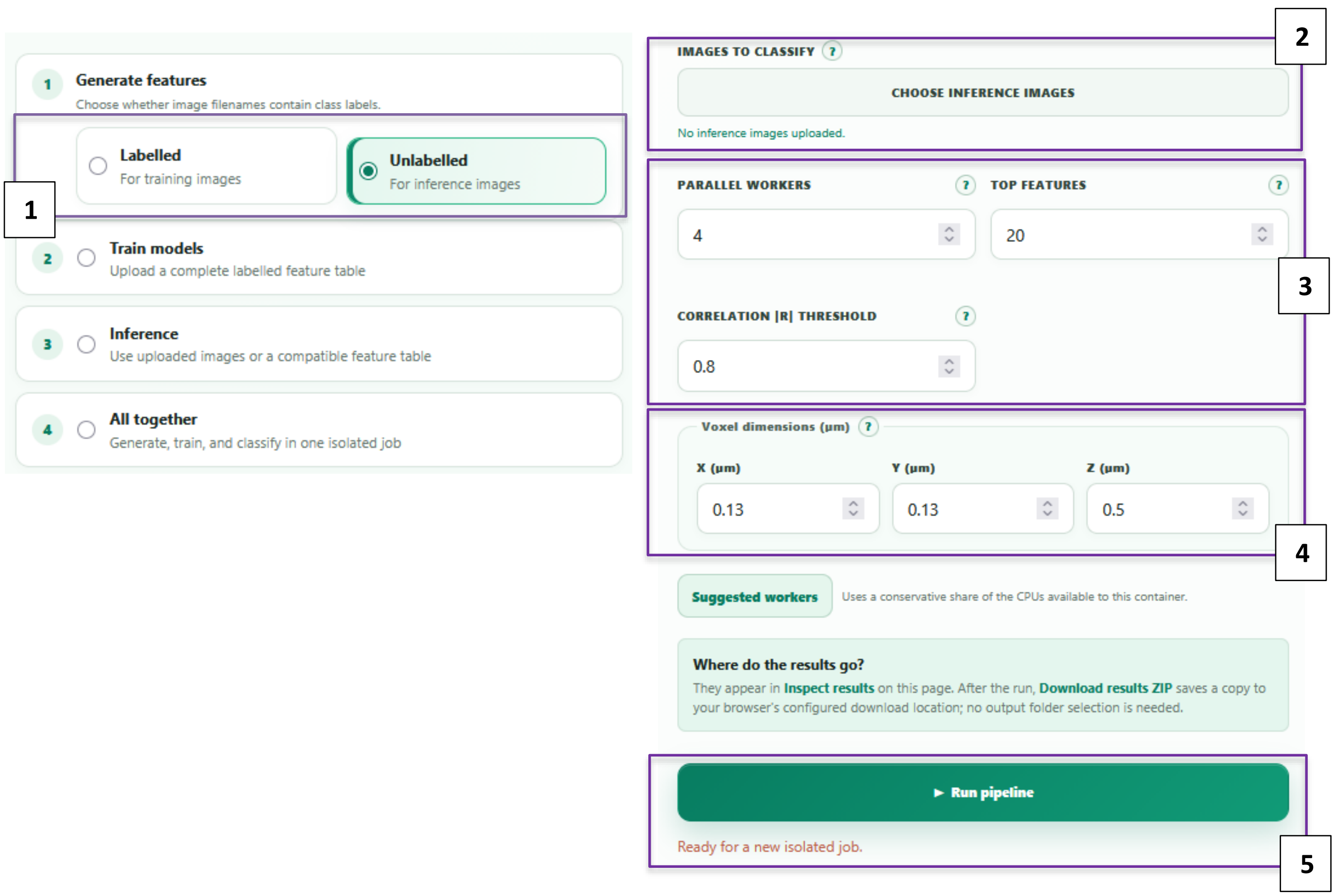
Inference

Inference consists of two steps: first, calculate features from the new images to obtain a list of features that the model can then use to make an inference on them, which is done in the second step.

Do step (1) — generate features from the new images:

1. Choose “Generate features” in “Unlabelled” mode. In unlabelled mode the labels are ignored because these images are the unknowns to be classified.
2. Point the software to the folder with the images for inference, using “Choose inference images”.
2. Set “Parallel workers” – how much of your CPU to dedicate to the analysis: full capacity when you are not otherwise using the computer, less if you are working on it at the same time. Suggested workers will provide optimal settings for running.
   “Top features” and “Correlation |R| threshold” do not matter here.
3. Adjust the voxel dimensions (the pixel size in micrometres) to match your images.
4. Click “Run pipeline”.

When the run starts, the software checks the image naming but does not read labels.



Do step (2) — run the model on those features:

1. Choose "Inference".
2. Point the software to the folder containing inference images using "Choose inference images" **OR** point the software to the feature table (the features you just generated in unlabelled mode) using "Choose feature table".
3. Point the software to the folder containing the trained model using the drop-down menu "Trained models".
4. All trained models are saved in the Docker folder, so you need to use the drop-down menu and choose the appropriate model. They are saved by the date when they were trained, followed by the number of models trained (one, when only one model was selected, or six, when all learners were selected), followed by a code created by Docker for tracking in case of problems. If you would like to delete an old model, or a model you ran as a test, select the model from the drop-down menu and click on "Delete selected job". You can also import trained models by using "Import _joblib files". It takes you to the folder with models; mark those models you would like to import (all or just one). Once they are imported, they will be used for inference.
5. Set "Parallel workers" as before. Suggested workers will provide optimal settings for running.
6. Click "Run pipeline".